DO NOW — RECALL

Unfamiliar programming challenges

Name: _____ Class: _____ Date: _____

1. What is a loop invariant?

2. What distinguishes a run from a distinct value?

3. Why handle empty arrays explicitly?

4. For a=[1,1,2,1], give count of distinct values and runs.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A property true before and after each completed iteration.

2. Model answer

Runs count contiguous groups; distinct counts globally unique values.

3. Model answer

Otherwise a[0] could be invalid and initial count wrong.

4. Model answer

Distinct values 2; runs 3 ([1,1],[2],[1]).

By the end, you can…

• A run is a maximal group of equal consecutive values, not a unique value across the whole array.

• Maintain invariant: after index i, the count equals the number of runs in the processed prefix.

• Handle empty input and test all-equal, alternating and separated-repeat arrays.

Vocabulary — know these terms

run — Consecutive equal-value group.

invariant — Property maintained across iterations.

edge case — Boundary or unusual input.

distinct — Different values irrespective of position.

NEW LEARNING

Read, annotate and explain

Understand

A run is a maximal group of equal consecutive values, not a unique value across the whole array.

Apply

Maintain invariant: after index i, the count equals the number of runs in the processed prefix.

Evaluate

Handle empty input and test all-equal, alternating and separated-repeat arrays.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace CountRuns on [4,4,5,5,5].

Predict first

Consecutive repeated fives belong to the same second run.

Explain and check

Explain the reason for the result.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace CountRuns for [3,3,4,4,3,5,5]. State where runs increment. [6]

Question 2

Write a valid C# CountRuns(int[] a) returning runs and handling empty input. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

For [1,2,1], compare runs and distinct counts, and assess time/extra-space complexity of CountRuns. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Runs always equal number of unique values.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.